This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document

REVIEWER ACTIVITIES: AIR FORCE - 99
DLA - IS

USER ACTIVITIES

DD FORM 672-1 (COORDINATED)

PROCUREMENT SPECIFICATION: MIL-B-3990

P.A. 11
Other Cust
Army - AT
Navy - AS

INTERNATIONAL INTEREST

SUPERSEDES:

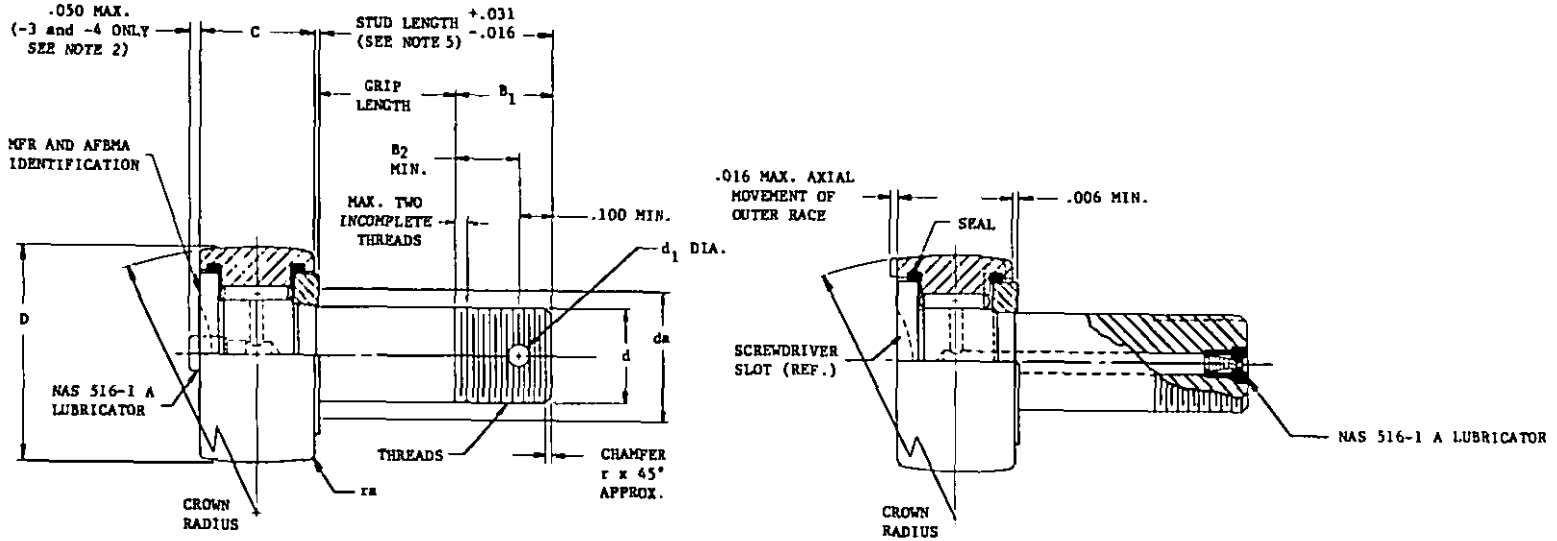
TITLE: BEARING, ROLLER, NEEDLE, TRACK ROLLER CROWN RADIUSED O.D. INTEGRAL STUD, TYPE VII ANTIFRICTION, INCH

MILITARY STANDARD

MS21447

SHEET 1 OF 2

Project No. 3110-0714



DIMENSIONS IN INCHES

FIRST DASH NO.	d STUD DIA.	D OUTER RING OUTSIDE DIA.	C OUTER RING WIDTH	CROWN RADIUS (REF)	THREAD SIZE UNJF-3A	B_1 THREAD LENGTH (REF)	da WASHER BACK-UP DIA. MIN.	B2 MIN.	d_1 COTTER PIN HOLE DIA.	ra 1/ FILLET MAX.	r	TOTAL RADIAL PLAY MAX.	LIMIT LOAD RATING lbf	2/ LB-IN MAX.	TRACK CAPACITY 60 HRC lbf	LOAD RATING AS A TRACK ROLLER lbf
-3	.1900	.5000	.281	10	.1900-32	.344	.297	.211	.070	.010	.031	.0017	790	8	385	395
-4	.2500	.6875	.281	10	.2500-28	.344	.359	.224	.076	.010	.031	.0017	940	20	525	470
-5	.3125	.7500	.344	10	.3125-24	.359	.422	.234	.076	.010	.047	.0017	1660	40	725	830
-6	.3750	.8750	.469	10	.3750-24	.359	.500	.265	.106	.025	.047	.0017	2720	55	1100	1360
-7	.4375	1.0000	.531	10	.4375-20	.422	.562	.283	.106	.025	.047	.0017	3860	155	1425	1930
-8	.5000	1.1250	.656	10	.5000-20	.422	.625	.314	.106	.040	.047	.0017	6080	205	1975	3040

FIRST DASH NO.	MASS (APPROX) lb
-3	.014 + (GRIP LENGTH NO. X .0005)
-4	.031 + (GRIP LENGTH NO. X .0009)
-5	.043 + (GRIP LENGTH NO. X .0014)
-6	.081 + (GRIP LENGTH NO. X .0020)
-7	.125 + (GRIP LENGTH NO. X .0026)
-8	.190 + (GRIP LENGTH NO. X .0035)

1/ THE CHAMFER ON BEARINGS MUST CLEAR THE MAXIMUM FILLET RADIUS GIVEN IN THE TABLE. THIS SPECIFICATION DOES NOT CONTROL BEARING CHAMFER CONTOURS.

2/ INSTALLATION TORQUE LUBRICATED THREADS.

FED. SUP CLASS 3110

APPROVED 12 SEP 84 REVISED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:40Z
Check the source to verify that this is the current version before use.

TABLE 1. TOLERANCE VALUES.

DIMENSIONS IN INCHES

D BASIC STUD DIA.		ALLOWABLE DEVIATION FROM d OF SINGLE MEAN DIA. d_{mp}		ALLOWABLE DEVIATION FROM COTTER PIN DIA. d_1	
OVER	INCL.	HIGH	LOW	HIGH	LOW
.1250	.5000	+0	-.0015	+.010	-0

D BASIC OUTER RING OUTSIDE DIA.		ALLOWABLE DEVIATION FROM D OF SINGLE MEAN DIA. D_{mp}		ALLOWABLE DEVIATION FROM OUTER RING WIDTH C	
OVER	INCL.	HIGH	LOW	HIGH	LOW
.4375	1.1250	+.0010	-.0005	+0	-.005

MATERIALS - STEEL, MIL-S-8690, MIL-E-7493, MIL-S-7420, ASTM A304 A322/A331, QQ-S-700, FED. STD. NO. 66, AISI/SAE 50100, 51100, AISI 862D, 8620H, AMS 6440, AMS 6442.

SEALS, ACETAL RESIN, NYLON, POLYESTER ELASTOMER, OR EQUIVALENT, CONSTRUCTION OPTIONAL.

FINISH - ROLLER O.D. AND SIDES CHROME PLATED IN ACCORDANCE WITH QQ-C-320, CLASS 2, EXCEPT THICKNESS TO BE .0004 - .0007 INCHES. EXPOSED SURFACES OF OTHER PARTS AS MOUNTED. CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE 1, CLASS 2, INCLUDING UNTHREADED PORTION OF THE STUD.

EXAMPLE OF PART NUMBER -

M S 2 0 A C = CAMFOLLOWER NEEDLE BEARING 1.1250 ROLLER, .500 STUD, SEALED, WITH CROWN RADIUS ON ROLLER O.D., LUBRICATION IN FLANGED END OF STUD, 1.250 GRIP LENGTH, 1.672 STUD LENGTH WITHOUT COTTER PIN HOLE, WITH MIL-G-81322 GREASE.

C = MIL-G-81322 GREASE, NO CODE = MIL-G-23827 GREASE.

A = NO COTTER PIN HOLE (NOT REQUIRED FOR TYPE T) OMIT FOR COTTER PIN HOLE.

GRIP LENGTH IN .0625 INCH INCREMENTS.

BEARING - F = LUBRICATOR IN FLANGED END OF STUD.
T = LUBRICATOR IN THREADED END OF STUD, NO COTTER PIN HOLE.
R = BEARING WITHOUT LUBRICATOR.

STUD DIAMETER IN .0625 INCH INCREMENTS.

NOTES:

1. THE COMPLETE MS IDENTIFICATION MAY BE OBTAINED BY ADDING SUFFIX CODE LETTERS TO THE BASIC MS NUMBER SHOWN IN THE EXAMPLE OF PART NUMBER.
2. MS21447-3 MAY BE CODED FOR RELUBRICATION FACILITIES IN THE FLANGED END ONLY. LUBRICATION FITTING FOR THE MS21447-3 MAY BE SHIPPED SEPARATELY AND INSTALLED ON ASSEMBLY. LUBRICATORS INSTALLED IN THE FLANGED END OF STUD ON MS21447-3 AND -4 MAY BE EXTENDED .05 IN. BEYOND THE FLANGED END THEREBY NECESSITATING AN INCREASE IN "W" OF .05 IN., ON ALL OTHER SIZES THE LUBRICATOR SHALL BE FLUSH OR INDENTED.
3. THE TRACK BRINELL CAPACITY IS CRITICAL IN RESPECT TO THE ROLLING CAPACITY OF THE BEARING. AN INCREASE IN HARDNESS OF THE TRACK WILL INCREASE THE BRINELL CAPACITY OF THE TRACK, BUT IN NO CASE SHOULD THE ROLLING CAPACITY OF THE BEARING BE EXCEEDED.
4. THE LIMIT LOAD RATING LISTED CAN BE DEFINED AS THE MAXIMUM LOAD WHICH CAN BE APPLIED TO A BEARING WITHOUT IMPAIRING THE SUBSEQUENT FUNCTIONING OF THE BEARING IN AIRFRAME APPLICATIONS. THE ULTIMATE OR STATIC FRACTURE LOAD RATING IS NOT LESS THAN 1.5 TIMES THE LIMIT LOAD RATING.
5. FOR INSPECTION PURPOSES NOMINAL STUD LENGTH IS THE SUM OF NOMINAL GRIP AND THE THREAD LENGTH (B_1 DIMENSIONS).
6. BEARINGS SHALL BE PREPACKED WITH GREASE AS CODED.
7. BEARINGS TO BE WRAPPED IN GREASE PROOF PAPER OR POLYETHYLENE BAG, INDIVIDUALLY PACKAGED AND MARKED WITH THE DATE OF LUBRICATOR.
8. ALL DIMENSIONS TO BE MET AFTER PLATING.

APPROVED 12 SEP 84 REVISED

USER ACTIVITIES:
REVIEWER ACTIVITIES: AIR FORCE - 99
DLA - IS
This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P. A. 11 Other Cust Army - AT Navy - AS	INTERNATIONAL INTEREST	TITLE: BEARING, ROLLER, NEEDLE, TRACK ROLLER CROWN RADIUSED O.D. INTEGRAL STUD, TYPE VII ANTIFRICTION, INCH	MILITARY STANDARD MS21447
PROCUREMENT SPECIFICATION MIL-B-3990	SUPERSEDES:		SHEET 2 OF 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:40Z
Check the source to verify that this is the current version before use.